

Caleb Hallinan

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EDUCATION

PhD in Biomedical Engineering

August 2023 – Present

Johns Hopkins School of Medicine, Baltimore, MD

Advisor: Jean Fan, PhD

Expected graduation: August 2027

Research focus: Deep learning for spatial transcriptomics, computational pathology, and multimodal omics analysis.

Bachelor of Arts in Statistics and Biology

August 2017 – May 2021

University of Virginia, Charlottesville, VA

RESEARCH EXPERIENCE

Biomedical Engineering PhD Candidate

August 2023 – Present

Johns Hopkins School of Medicine

Advisor: Jean Fan, PhD

- Develop deep learning and computational biology methods for analyzing spatial transcriptomics, histopathology images, and multimodal molecular datasets.
- Utilize deep learning models to predict spatial gene expression from histology images using representations learned from tissue morphology.
- Developed Off-target Probe Tracker, a computational method to detect off-target probe binding in 10x Genomics Xenium spatial transcriptomics panels.

Research Assistant II

September 2021 – June 2023

Boston Children's Hospital & Harvard Medical School

Advisor: Kwonmoo Lee, PhD

- Developed a heterogeneity-preserving feature selection algorithm for subtype discovery across gene, protein, and image-derived feature spaces.
- Applied machine learning and image analysis methods to live-cell imaging datasets of breast cancer stem cells to quantify morphology, motility, and phenotypic heterogeneity.
- Used deep learning algorithms for cell segmentation and feature extraction from microscopy image datasets.
- Managed Linux research servers and supported computational infrastructure for laboratory research workflows.
- Presented recent research papers and methodological developments to lab members through journal clubs and internal research meetings.

Undergraduate Research Assistant

May 2020 – August 2021

Focused Ultrasound Foundation & University of Virginia

Advisors: Frederic Padilla, PhD & Tianxi Li, PhD

- Performed statistical analysis of flow cytometry data to evaluate the effects of focused ultrasound treatment on mouse tumor models.
- Processed, analyzed, and visualized brain tumor datasets to distinguish normal brain matter from tumor tissue.
- Translated the R package 'randnet' into Python, porting statistical network analysis functionality into a Python-based implementation.

PUBLICATIONS

Peer-Reviewed Journal Articles:

- J6. C. Hallinan**, H.J. Ji, S.L. Salzberg, J. Fan. “Evidence of off-target probe binding affecting 10x Genomics Xenium gene panels compromise accuracy of spatial transcriptomic profiling.” *eLife*, 2026.
- J5.** H. Zhou, P. Panwar, B. Guo, **C. Hallinan**, S. Ghazanfar, S. C. Hicks. “Spatial mutual nearest neighbors for spatial transcriptomics data.” *Bioinformatics*, 2025.
- J4. C. Hallinan***, A. Abul-Basher*, and K. Lee. “Heterogeneity-Preserving Feature Selection for Subtype Discovery.” *Nature Communications*, 2025.
- J3.** S. Busatto, T. Song, H. J. Kim, **C. Hallinan**, ..., K. Lee, M. Moses. “Breast Cancer-Derived Extracellular Vesicles Modulate the Cytoplasmic and Cytoskeletal Dynamics of Blood-Brain Barrier Endothelial Cells.” *Journal of Extracellular Vesicles*, 2025.
- J2.** J. Jang, Y. Kim, B. Westgate, Y. Zong, **C. Hallinan**, A. Akalin, and K. Lee. “Screening Adequacy of Unstained Fine Needle Aspiration Samples Using a Deep Learning-based Classifier.” *Scientific Reports*, 2023.
- J1.** J. Jang, **C. Hallinan**, and K. Lee. “Protocol for live cell image segmentation to profile cellular morphodynamics using MARS-Net.” *STAR Protocols*, 2022.

Manuscripts Under Review:

- P2.** D. Velazquez, **C. Hallinan**, R. An, K. Clifton, J. Fan. “Spatial Transcriptomics As Rasterized Image Tensors (STARIT) characterizes cell states with subcellular molecular heterogeneity.” 2026.
- P1.** **C. Hallinan**, C. H. G. Lucas, J. Fan. “Impact of Data Quality on Deep Learning Prediction of Spatial Transcriptomics from Histology Images.” 2025.

*Equal Contributors

PRESENTATIONS

Oral Presentations:

- O6.** “Understanding Spatial Transcriptomics and Evaluating Probe Accuracy in 10x Genomics Xenium Technology,” Models, Inference & Algorithms (MIA) Seminar, Broad Institute, Cambridge, MA, 2026.
- O5.** “Predicting 3D Molecular Maps from 2D Histology using Deep Learning,” Genomics Joint Lab Meeting, Johns Hopkins University, Baltimore, MD, 2026.
- O4.** “Impact of Data Quality on Gene Expression Prediction using Deep Learning on Histopathology Images,” Genomics Joint Lab Meeting, Johns Hopkins University, Baltimore, MD, 2025.
- O3.** “Deconvolution of Cellular Heterogeneity for Sub-Type Discovery by Analyzing Feature Variation,” Vascular Biology Program Work in Progress, Boston Children’s Hospital & Harvard Medical School, Boston, MA, 2022.
- O2.** “Machine Learning Approaches Applied to the Prediction of COVID-19 Spread and Cell Motility Phenotyping,” Vascular Biology Program Work in Progress, Boston Children’s Hospital & Harvard Medical School, Boston, MA, 2022.
- O1.** “Ultrasound Microbubble Tumor Analysis,” Focused Ultrasound Foundation Summer Intern Presentations, Focused Ultrasound Foundation, Charlottesville, VA, 2021.

Poster Presentations:

- P3.** “Impact of Data Quality on Deep Learning Prediction of Spatial Transcriptomics from Histology Images,” AI in Molecular Biology Keystone Symposia, Eldorado Hotel & Spa, Santa Fe, NM, 2025.
- P2.** “Phenotyping of Heterogeneous Live Cell Motility and Morphology,” Dr. M. Judah Folkman Research Day, Boston Children’s Hospital & Harvard Medical School, Boston, MA, 2023.
- P1.** “Deep-Hetero: A Deep Metric Learning with UMAP-based Clustering Approach for Identifying Heterogeneity in Cells,” Dr. M. Judah Folkman Research Day, Boston Children’s Hospital & Harvard Medical School, Boston, MA, 2022.

TEACHING AND MENTORSHIP

Research Mentor, High School Student

June 2026 – Present

Whiting School of Engineering Summer Internship Program (WISE), Johns Hopkins University

- Mentor a high school student interested in biomedical engineering, computational biology, and data-driven biomedical research.

- Introduce the student to core concepts in Python programming, biomedical image analysis, and spatial omics through structured research discussions and hands-on exercises.
- Provide guidance on scientific communication, research expectations, and academic development.

Research Mentor, Undergraduate Student

May 2025 – Present

Biomedical Engineering Department, Johns Hopkins University

- Mentor an undergraduate student in computational biology, spatial transcriptomics, histopathology image analysis, and Python-based research workflows.
- Support student development through regular meetings, technical feedback, and project-specific troubleshooting.

Course Instructor, Neural Networks from Scratch

January 2026

Biomedical Engineering Department, Johns Hopkins University

- Designed and taught a hands-on course on building neural networks from first principles.
- Developed lectures and coding tutorials covering core deep learning theory, backpropagation, convolutional neural networks, Transformers, and PyTorch implementations.
- Guided students through practical neural network implementation exercises emphasizing mathematical intuition and reproducible coding practices.

Course Instructor, Deep Learning for Spatial Transcriptomics

August 2025 – November 2025

Biomedical Engineering Department, Johns Hopkins University

- Designed and taught an original course on deep learning methods for spatial transcriptomics.
- Created conceptual lectures and hands-on PyTorch tutorials covering representation learning, image-based prediction, spatial omics data structures, and model evaluation.
- Supported student learning through coding demonstrations, discussion-based instruction, and individualized project guidance.

Teaching Assistant, Genomic Data Visualization

January 2025 – March 2025

Biomedical Engineering Department, Johns Hopkins University

- Supported student learning by attending lectures, answering questions, and providing individualized guidance during office hours.
- Evaluated student progress by grading assignments and providing constructive feedback on genomic data visualization techniques.
- Assisted students with computational approaches for visualizing and interpreting high-dimensional genomic datasets.

Teaching Assistant, Data Science for Public Health I/II

January 2024 – May 2024

Biostatistics Department, Johns Hopkins Bloomberg School of Public Health

- Facilitated student understanding of data science concepts through office hours, assignment support, and grading.
- Provided feedback on computational assignments involving data analysis, statistical reasoning, and reproducible programming practices.

Teaching Assistant, Computational Analysis of Heterogeneity in Cellular Images Nano Course

January 2023

Harvard Medical School Curriculum Fellows Program, Harvard Medical School

- Guided 20 participants through hands-on analysis of live-cell image datasets.
- Supported instruction on core algorithms for cell segmentation, edge detection, and cell tracking.
- Assisted participants in applying computational image analysis methods to quantify cellular heterogeneity.

Teaching Assistant, Regression Analysis

January 2020 – May 2021

Statistics Department, University of Virginia

- Supported instruction for 80 students in a SAS-based, project-oriented regression analysis course.

- Held office hours, graded assignments, answered in-class questions, and helped students develop statistical modeling skills.
- Provided individualized feedback on regression projects and statistical programming assignments.

Teaching Assistant, Introduction to Chemistry Lab
Chemistry Department, University of Virginia

September 2019 – December 2019

- Supervised 24 students in weekly chemistry laboratory sessions.
- Designed presentations, graded assignments, held office hours, and supported safe laboratory practices.

LEADERSHIP, SERVICE, AND OUTREACH

Mentor, Hopkins Biomedical Engineering Application Assistance Program (BMEAAP) *November 2023 – Present*
Biomedical Engineering Department, Johns Hopkins University

- Mentor underrepresented applicants to the Johns Hopkins Biomedical Engineering PhD program.
- Provide feedback on CVs, personal statements, and application materials.
- Conduct mock interviews and advise applicants on graduate school preparation and research communication.

Co-President, Johns Hopkins Biomedical Engineering Student Council

September 2023 – Present

Biomedical Engineering Department, Johns Hopkins University

- Serve as co-President representing over 200 doctoral students across Johns Hopkins Biomedical Engineering.
- Coordinate council meetings, lead graduate student initiatives, and advocate for student needs.
- Serve as a liaison between graduate students, faculty, and administration to improve the doctoral student experience.

HONORS AND AWARDS

- Graduated with Distinction, University of Virginia
- Dean's List, University of Virginia. 4 of 5 eligible semesters.

May 2021

January 2018 – December 2019

TECHNICAL SKILLS

- Programming: Python, R, SQL, Bash, MATLAB, SAS, Mathematica, C, HTML, CSS
- Machine Learning and AI: PyTorch, TensorFlow, deep learning, convolutional neural networks, Transformers, representation learning, deep metric learning, feature selection, image segmentation, model evaluation
- Computational Biology: spatial transcriptomics, single-cell RNA-seq, omics analysis, histopathology image analysis, cell biology, cell segmentation, feature extraction, ImageJ
- Data Science and Statistics: statistical analysis, regression modeling, clustering, dimensionality reduction, data visualization, high-dimensional data analysis
- Research Computing: Linux server management, shell scripting, Git, reproducible computational workflows, LaTeX
- Productivity and Communication: scientific writing, technical teaching, oral presentation, mentorship

REFERENCES

References available upon request.